Ganesh Gupta

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Experience

Oracle | *Software Engineer, AI Platform* Jun 2025 – Present

- Built a **distributed executor** in **Python** and **Java** that dispatches workloads to **Docker** or **AWS Lambda** via typed **REST APIs**, holding **sub-200ms p99** across **millions of daily transactions**.
- Designed a content-addressed **S3** artifact store with memoization and per-step **Redis** checkpointing, so interrupted workflows **resume from the last committed step** at **99%+ uptime**.
- Built **distributed search and retrieval** infrastructure on **Elasticsearch** with knowledge-graph enrichment, serving indexed queries behind the same typed API layer.
- Wrote **fault-tolerant** scheduling and placement logic for shared multi-cloud compute, keeping workloads running through **node failures** and capacity limits.

QureAI | *AI Engineer* Mar 2026 – Present

- Cut **p99 latency 66% (420ms to 140ms)** across **50K+ daily requests** by moving synchronous serving to **async processing** with horizontal **autoscaling**.
- Built **PySpark** jobs and **Airflow** DAGs for large-scale data preparation, and **CI/CD** with quality gates and **automated rollback**.
- Raised **top-3 retrieval recall from 61% to 89%** on a 2K-query eval set by rebuilding the index with **hybrid dense-sparse retrieval**.

Nomura Research Institute Financial Technologies | *Software Engineer* Sep 2019 – Aug 2023

- Sustained **99.9% uptime with zero data loss** for **1M+ users** by architecting **25+ Java Spring Boot microservices** for financial transaction processing on **AWS (EC2, RDS, SQS)**.
- Cut settlement-blocking nightly aggregation **81% (8 minutes to 90 seconds)** by redesigning **SQL and PL/SQL** with **indexed views** and **partitioning** to remove row-level **lock contention**.
- Cut production escalations **60%** in one quarter by standardizing retry, timeout, and failure contracts across **25+ services** and driving **root cause analysis**.
- Led **architecture and code reviews** across production services, mentoring engineers on **system design**, reliability, and engineering standards.

Projects

Distributed NoSQL Database `github.com/ganeshhgupta/distributed-kv-store`

- **Fault-tolerant distributed storage** that confirms writes only after a **majority of nodes** store them, recovers crashed nodes from a **write-ahead log**, and resyncs partitioned replicas via **leader-follower replication**.

RootCause AI `github.com/ganeshhgupta/rootcause-ai`

- **LangGraph** multi-agent system that triages production alerts, routes each to a specialized sub-agent with **tool use**, and writes a structured **root-cause brief**.

Watch-your-LLM `github.com/ganeshhgupta/watch-your-llm`

- Observability platform that traces model calls, latency, and token usage, with a **FastAPI** backend, **PostgreSQL**, and a **React/TypeScript** dashboard.

Education

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0* Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science* Aug 2016 – Jul 2020

Skills

Languages: Java, Python, SQL, TypeScript

Backend and Distributed Systems: Spring Boot, FastAPI, REST APIs, Microservices, Kafka, SQS, Redis, Replication, Fault Tolerance, System Design

Data and Storage: PostgreSQL, Oracle, Elasticsearch, S3, PySpark, Airflow

Cloud and Reliability: Azure, AWS, Docker, Kubernetes, Terraform, Prometheus, Grafana, CI/CD with Automated Rollback

Certifications: AWS Certified Developer Associate, GCP Professional MLOps Engineer